Collaborative Discussion 2: Agent Communication Languages

This discussion will last for 3 weeks, starting in Unit 5 and ending in Unit 7. Ensure you include appropriate citations and references in all your posts.

Discussion Topic: Agent Communication Languages

What are the potential advantages and disadvantages of the use of agent communication languages such as KQML? How do they compare with method invocation in Python or Java?

Learning Outcomes

- An understanding of the main agent models in use today and their grounding in artificial intelligence research.
- The knowledge and skills required to develop, deploy and evaluate the tools and techniques of intelligent systems to solve real-world problems.
- An understanding of contemporary research issues in the area of intelligent agent systems.

Assignment Guidance

- Your initial posting should respond to the question and be at least 200 words long. This should be labelled as '**Initial Post**'
- You will then respond to **at least 2 of your peers' posts** in Unit 6 (each labelled as '**Peer Response**'). To guide your responses, look at the guidelines for the peer review process on the **Department's homepage**. Focus on the possible measures that could have been put in place in order to prevent the incidents highlighted by your peers. Please try to limit your response posts to **200-300 words maximum**, so that others may be encouraged to reflect on, and respond to your ideas.
- In Unit 7, you should provide a summary post based on your initial post, the feedback from your peers and the content of the three units. Please label this as '**Summary Post**'. It should be **300 words**.

- Referencing: When you have referred to other authors thoughts, ideas and opinions in your posts, you must reference the author as you would in an academic assignment using the UoEO Harvard reference style.
- You will be not be assessed on your contribution to this forum throughout the module, but the tutor will post group feedback via a module announcement.
- This activity forms a component of your e-portfolio which you will submit in Unit 12. All e-portfolio activities are intended to demonstrate your ability and strengths through evidence and reflection.

Initial Post

by Yasmin Almeida Ferrera - Friday, 28 August 2026, 10:38 AM
Number of replies: 3

Agent Communication Languages (ACLs), such as KQML, were developed to allow autonomous agents to communicate through messages representing actions such as requests, questions and information sharing (Labrou and Finin, 2000). This is fundamentally different from method invocation in Python or Java. When a Python or Java programme calls a method, the developer normally knows exactly which operation is being requested, what inputs it requires and what type of result should be returned. With KQML, the focus is instead on communicating an intention, allowing the receiving agent to decide how to respond.

This is where I see both the main strength and weakness of ACLs. They can make a system much more flexible because agents do not need to know the internal implementation of other agents. This is particularly useful when agents are independent or distributed. However, the same autonomy can make the system harder to control. Communication requires additional processing, and the outcome may not be as predictable as a direct method call.

Python and Java therefore have an important advantage in terms of control and reliability. A method has a defined interface, making it easier to test, debug and predict its behaviour. The limitation is that the components become more dependent on each other. If the method changes, the code relying on it may also need to change. For a multi-agent

system where components need to operate independently, this type of dependency can become a limitation.

There is also a deeper problem: communication does not necessarily equal understanding. Two agents can follow the same protocol and still interpret information differently. This is why semantic interoperability is important in systems where agents need to share knowledge, not just exchange messages (Malik and Malik, 2018).

I find this particularly relevant to current AI agents. MCP provides a standardised way for AI applications to connect with tools and external resources (Anthropic, 2024), while multi-agent LLM research shows that agents can communicate to challenge and improve each other's reasoning (Du et al., 2024). In this sense, the idea behind ACLs is still relevant, even if the technology has changed.

For me, the interesting trade-off is therefore not simply KQML versus Python or Java. It is control versus autonomy. Method invocation gives us predictable behaviour, while agent communication gives agents more freedom to cooperate and adapt. The question is whether that additional autonomy actually produces better results, or whether the loss of predictability becomes too high a price to pay.

References

Anthropic (2024) *Model Context Protocol specification*. Available at: **https://modelcontextprotocol.io/introduction**(Accessed: 25 August 2026).

Du, Y., Li, S., Torralba, A., Tenenbaum, J.B. and Mordatch, I. (2024) 'Improving factuality and reasoning in language models through multiagent debate', *Proceedings of the 41st International Conference on Machine Learning*, 235. Available at: **https://openreview.net/pdf?id=zj7YuTE4t8N** (Accessed: 25 August 2026).

Labrou, Y. and Finin, T. (2000) *Agent Communication Languages: Past, Present and Future*. Tutorial presented at ICMAS 2000, Boston, July 2000. Available at: **https://userpages.cs.umbc.edu/finin/talks/icmas00.pdf** (Accessed: 28 August 2026).

Malik, M. and Malik, S.K. (2018) 'Semantic web as the next generation smart web', *Journal of Emerging Technologies and Innovative Research*, 5(6), pp. 580–594. Available at: **https://www.jetir.org/papers/JETIR1806326.pdfR** (Accessed: 28 August 2026).

In reply to Yasmin Almeida Ferrera

Peer Response

by Lara Koller - Saturday, 29 August 2026, 10:11 PM

Hi Yasmin,

I particularly liked your point that communication does not necessarily equal understanding. This seems especially important for ACLs because standardising a message does not guarantee that two agents interpret its content in the same way. KQML uses performatives such as *tell* and *ask-if* to represent communicative intention, while FIPA-ACL includes acts such as *inform* and *request* (Wooldridge, 2009; Tsiakas and Murray-Rust, 2024). FIPA-ACL's semantics also refer to internal states such as beliefs and intentions. Wooldridge (2009) argues that this creates a semantic conformance problem because it can be difficult to verify externally whether an implemented agent actually satisfies the conditions associated with a communicative act. A correctly structured message therefore does not by itself establish that the sender satisfies its intended semantics.

Your framing of control versus autonomy also made me consider whether the loss of predictability comes from autonomy itself, or from how that autonomy is constrained. Singh, Christie and Chopra (2024) complicate this with Langshaw. Their approach combines flexibility with formal semantics and procedures for determining protocol safety and liveness, while supporting the generation of message-oriented protocols suitable for flexible asynchronous enactment. This suggests that autonomy and control need not be opposites.

I would therefore shift the trade-off from control versus autonomy towards flexibility versus verifiability: how much freedom can agents retain while still allowing us to specify and verify acceptable interactions?

Your MCP comparison also raises a useful distinction. MCP standardises how AI applications connect to external systems, including tools and data sources (Anthropic, 2024), whereas ACL performatives represent communicative acts between agents. Do you see MCP as continuing the ACL idea, or as solving a different interoperability problem?

References

Anthropic (2024) *Introducing the Model Context Protocol*. Available at: **https://www.anthropic.com/news/model-context-protocol** (Accessed: 29 August 2026).

Singh, M. P., Christie V., S. H. and Chopra, A. K. (2024) 'Langshaw: Declarative Interaction Protocols Based on Sayso and Conflict', *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24)*, pp. 202-210. Available at: **https://doi.org/10.24963/ijcai.2024/23**

Tsiakas, K. and Murray-Rust, D. (2024) ‘Unpacking Human-AI interactions: From Interaction Primitives to a Design Space’, *ACM Transactions on Interactive Intelligent Systems*, 14(3), pp. 1-51. Available at: **https://doi.org/10.1145/3664522**

Wooldridge, M. (2009) *An Introduction to MultiAgent Systems*. 2nd edn. Chichester: John Wiley & Sons.

In reply to Yasmin Almeida Ferrera

Peer Response

by Lana Menezes Fidelis - Tuesday, 1 September 2026, 10:31 PM

Hello Yasmin! Framing the trade-off as control versus autonomy is a clear way to cut through the KQML-versus-method-call comparison, and connecting it to MCP and multi-agent debate research (Du et al., 2024) makes a strong case that the underlying design question has outlived the original ACL technology.

I would push a little further on the "communication does not equal understanding" point. This is essentially the symbol grounding problem in a distributed setting: even with a shared protocol like KQML, performatives only specify the type of speech act, not the meaning of the terms inside the message content (Chaib-draa and Dignum, 2002). FIPA-ACL tried to address part of this by requiring agents to share content languages and ontologies explicitly rather than assuming them, though as your Malik and Malik (2018) reference suggests, semantic web style ontology alignment does not fully eliminate the risk of divergent interpretation. Wooldridge (2009) makes a related point that agent communication only works reliably when agents share a common mental model of the world, something that cannot be guaranteed simply by agreeing on a message syntax.

On the reliability point, I think there is a useful middle ground worth naming: many practical multi-agent systems, and MCP-based tool use is a good example, keep the transport and message structure standardised and machine-checkable (closer to a method call's defined interface) while leaving the content of the request open to interpretation by the receiving model. This partially recovers some of the testability you attribute to Python or Java, while still preserving the flexibility ACLs were designed for. It suggests the real design choice is not fully autonomous versus fully deterministic, but where along that spectrum the interface boundary is drawn.

This links to Jennings' (2000) argument that coordination in multi-agent systems requires an explicit trade-off between the flexibility of negotiation-based interaction and the predictability needed for dependable engineering, a tension that maps closely onto your control-versus-autonomy framing. More recent work on LLM-based agents (Guo et al., 2024) also suggests that as natural language itself becomes the communication medium, the boundary between "message" and "method call" is blurring further, since a natural language request can carry both structured intent and open-ended content within a single exchange.

References

Guo, T., Chen, X., Wang, Y., Chang, R., Pei, S., Chawla, N.V., Wiest, O. and Zhang, X. (2024) 'Large language model based multi-agents: a survey of progress and challenges', in *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24)*, pp. 8048-8057.

Jennings, N.R. (2000) 'On agent-based software engineering', *Artificial Intelligence*, 117(2), pp. 277-296.

Wooldridge, M. (2009) *An Introduction to MultiAgent Systems*. 2nd edn. Chichester: Wiley.

In reply to Yasmin Almeida Ferrera

Re: Initial Post

by Yasmin Almeida Ferrera - Tuesday, 8 September 2026, 4:48 PM

Summary Post

The discussion has highlighted that the main difference between Agent Communication Languages (ACLs) and traditional method invocation is not simply how messages are structured, but how much freedom is given to the receiving agent. My initial argument framed this as a trade-off between control and autonomy. However, the responses from Lara and Lana suggest that a more precise way of describing this tension is "flexibility versus verifiability".

Lara highlighted an important limitation of ACLs: standardising communication does not guarantee that agents interpret messages in the same way. KQML and FIPA-ACL can define communicative intentions such as tell, ask and request, but verifying whether an agent actually satisfies the semantic conditions associated with these acts can be difficult (Wooldridge, 2009). Lana extended this argument by connecting it to the symbol grounding problem. Even when agents share a communication protocol, they may interpret the concepts contained within messages differently unless they also share appropriate ontologies and assumptions (Chaib-draa and Dignum, 2002).

At the same time, both responses challenged the idea that autonomy necessarily reduces control. Research such as Langshaw demonstrates that flexible agent interactions can be combined with formal mechanisms for checking protocol safety and liveness (Singh, Christie and Chopra, 2024). This suggests that autonomy and reliability do not have to be mutually exclusive; instead, system designers need to determine where flexibility is acceptable and where interactions require stronger verification.

This perspective also changes how I view modern systems such as MCP. Rather than being a direct continuation of ACLs, MCP addresses a related but distinct interoperability problem by standardising how AI applications interact with tools and external resources (Anthropic, 2024). Similarly, LLM-based multi-agent systems increasingly blur the boundary between structured messages and natural-language requests (Guo et al., 2024).

Overall, the discussion has reinforced that effective multi-agent communication requires more than a shared protocol. It requires balancing flexibility, semantic alignment and verifiability according to the reliability requirements of the system.

References

Anthropic (2024) Introducing the Model Context Protocol. Available at: https://www.anthropic.com/news/model-context-protocol (Accessed: 29 August 2026).

Chaib-draa, B. and Dignum, F. (2002) ‘Trends in agent communication language’, Computational Intelligence, 18(2), pp. 89–101.

Guo, T., Chen, X., Wang, Y., Chang, R., Pei, S., Chawla, N.V., Wiest, O. and Zhang, X. (2024) ‘Large language model based multi-agents: a survey of progress and challenges’, in Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24), pp. 8048–8057.

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Wooldridge, M. (2009) An Introduction to MultiAgent Systems. 2nd edn. Chichester: Wiley.